

Derbyshire Shared Care Pathology Guidelines

Diagnosis of Polycystic Ovarian Syndrome in adult females

Background

- Estimated prevalence of PCOS is 5-20% in reproductive aged women
- Risk factors for PCOS include family history of PCOS, Obesity, type 2 diabetes
- PCOS may present clinically with menstrual irregularities, signs of androgen excess (e.g. hirsutism and acne) and reduced fertility
- It is associated with increased risk of type 2 diabetes, dyslipidaemia and cardiovascular events

PCOS is a diagnosis of exclusion and other conditions that cause menstrual irregularities and androgen excess should be adequately ruled out before it is safe to accept a diagnosis of PCOS. Clinical and biochemical features of related disorders are detailed in Appendix 1.

Diagnostic criteria - (International PCOS Network (2023))

Diagnosis and treatment of polycystic ovary syndrome (PCOS) has been controversial, with challenges defining individual components within the diagnostic criteria and significant clinical heterogeneity, which is further influenced by ethnic differences and changes in clinical features across the life course.

After investigations and the exclusion of differential diagnoses consider a diagnosis PCOS in adults (> 3 years post menarche to perimenopause) if a person has two of the following:

- Irregular cycles and ovulatory dysfunctions (eg cycles of > 35 or < 21 days or < 8 cycles per year that continue for more than two years).
- Hyperandrogenism:
 - o EITHER Biochemically
 - Raised testosterone or free testosterone or if these are normal then raised androstenedione or DHEAS levels.
 - o OR Clinically
 - modified Ferriman Gallwey score (mFG) ≥ 8
- Ultrasound for polycystic ovarian morphology
 - o Follicle number per ovary (FNPO) ≥ 20 in at least one ovary
 - o Ovarian volume (OV) ≥ 10 ml or follicle number per section (FNPS) ≥ 10 in at least one ovary

Initial investigations

- **Clinical History and Clinical Examination**
 - o A diagnosis of PCOS can be made if Irregular cycles /ovulatory dysfunctions AND hirsutism with a modified Ferriman Gallwey score (mFG) ≥ 8 is present AND alternative cause of symptoms has been excluded (see Appendix 1)
- **Testosterone and SHBG** (to confirm biochemical hyperandrogenaemia)
 - o If the testosterone is ≥ 4 nmol/L then please send Advice and Guidance to Endocrinology
 - o If the testosterone level is >2.9 nmol/L when analysed by immunoassay at RDH and CRH, the laboratory confirm the result and exclude assay interference using a mass-spectrometry method ('T-MS Testosterone').
 - o For QHB testosterone results please check for comments regarding sending specific further request for T-MS testosterone where needed.
 - o (Whilst Oestrogen within COC may affect accurate measurement of biochemical hyperandrogenism, we do not routinely recommend stopping COC)
- Consider measuring the following, particularly if the patient presents with menstrual disturbance:
 - o **TSH** (to exclude hypothyroidism)
 - o **FSH** (to exclude premature ovarian failure)
 - o **Prolactin** (to exclude hyperprolactinaemia)
 - if >600 mIU/L on two separate occasions then please send Advice and Guidance to Endocrinology.
- Pelvic ultrasound for PCOS diagnosis is only needed where the diagnosis cannot be made following the above clinical and biochemical assessments. However, ultrasound may also be important to exclude other pathology including ovarian. Assessment of endometrium by ultrasound can also be important. Ultrasound requests will be considered depending on clinical detail provided.

Note: AMH levels may have a role in PCOS diagnosis in some patients but would be considered as part of any secondary care investigations

Second line tests

Exclusion of other causes of hyperandrogenaemia is dependent on clinical symptoms (see appendix 1). This should include consideration in rapid virilisation. Referral to endocrinology may be appropriate in these cases.

Investigation of long-term health consequences of PCOS

Once a patient is diagnosed with PCOS, further investigations may be required. PCOS is associated with an increased risk of type 2 diabetes, depression and cardiovascular disease. It is therefore suggested that the patient have a baseline HbA1c, blood pressure and lipid profile measured, alongside a depression screen and then their cardiovascular risk scores rechecked at appropriate intervals.

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131 (24/7)
Endocrinology Advice	01332 783284

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271

Burton

Duty Biochemist	01283 593313 (8am - 4pm Mon-Fri)
On Call Consultant Biochemist	Via QHB switchboard, 01283 511511 (24/7)

Patient information website:

<https://www.nhs.uk/conditions/polycystic-ovary-syndrome-pcos/>

References

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2. Royal College of Obstetrics and Gynaecologists Guideline No. 33, 2007. Long-term consequences of Polycystic Ovary Syndrome.
3. Piltonen, Terhi T et al. "Validation of an Anti-Müllerian Hormone Cutoff for Polycystic Ovarian Morphology in the Diagnosis of Polycystic Ovary Syndrome in the HARMONIA Study: Protocol for a Prospective, Noninterventional Study." *JMIR research protocols* vol. 13 e48854. 6 Feb. 2024,
4. Wild, R.A. et al., 2010. Assessment of Cardiovascular Risk and Prevention of Cardiovascular Disease in Women with the Polycystic Ovary Syndrome: A Consensus Statement by the Androgen Excess and Polycystic Ovary (AE-PCOS) Society. *JCEM*, 95(5), pp.2038-2049
5. Elecys AMH assay Product Information 2022-01 V6.0 English. Roche Diagnostics GmbH
6. Dietz de Loos A, Hund M, Buck K, et al. Antimüllerian hormone to determine polycystic ovarian morphology. *Fertil Steril* 2021;116(4):1149-1157.(APHRODITE study)

Document title	Diagnosis of polycystic Ovarian syndrome in Adult Females		
Revision	5	Document Reference	CHISCP30
Reason for Review	Routine Review/Update to 2023 guidelines		
Review history	Version/Date	Authors	Reason for review
	V1 Sept 2013	Dr L Hawke Dr R Stanworth, Dr P Masters, Dr P Blackwell	New Document
	V2 Nov 2018	Dr R Stanworth, Dr P Blackwell Mrs H Seddon	Routine Review
	V3 Nov 2024	Dr R Stanworth, Dr P Blackwell Ms S Knowles	Extension
	V4 April 2025	Dr R Stanworth, Dr P Blackwell Ms S Knowles	Extension
Approval	Date	Name(s)	Designation
Secondary Care UHDB	13.06.25	Dr R Stanworth	Consultant Endocrinologist
	13.06.25	J Weston	General Ultrasound Clinical Manager
Secondary Care CRH	05.08.25	Dr R Robinson	Consultant Endocrinologist
Primary Care	07.08.25	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	11.07.25	Ms S Knowles	Consultant Clinical Scientist

Appendix 1.
Exclusion of related disorders (taken from Ehrmann, 2005)

Condition	Presence of hyperandrogenism	Oligo- or amenorrhoea	Clinical features	Biochemical features
Non-classical CAH (21-hydroxylase deficiency)	Yes	Not often	Family history of infertility, hirsutism, common in Ashkenazi Jews	Elevated (morning, basal) 17-hydroxyprogesterone or after Synacthen stimulation
Cushing's syndrome	Yes	Yes	Hypertension, striae, easy bruising	Elevated 24-hour urinary cortisol
Hyperprolactinaemia	None or mild	Yes	Galactorrhoea	Elevated serum prolactin
Primary hypothyroidism	None or mild	May be present	Goitre may be present	Elevated serum TSH
Acromegaly	None or mild	Often	Acral enlargement, coarse features, prognathism	Increased plasma IGF-1
Premature ovarian failure	None	Yes	May be associated with other auto-immune endocrinopathies	Elevated serum FSH
Simple obesity	Often	Not often	Exclusion	None
Virilising adrenal or ovarian neoplasm	Yes	Yes	Clitoromegaly, extreme hirsutism, male pattern alopecia	Extremely elevated plasma androgen level
Drug-related condition	Often	Variably	Drug history: Use of androgens, valproic acid, cyclosporin, spironolactone	None